

Name: _____ Enrollment No: _____ Marks: 10

Question 1 [3 × 2 = 6 marks]

1. For married couples living in a certain suburb, the probability that the husband will vote on a bond referendum is 0.21, the probability that the wife will vote on the referendum is 0.28, and the probability that both the husband and the wife will vote is 0.15. What is the probability that
 - a) at least one member of a married couple will vote?
 - b) a wife will vote, given that her husband will vote?
 - c) a husband will vote, given that his wife will not vote?
2. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Question 2 [4 × 1 = 4 marks]

For the following frequency distribution:

Class interval	0–10	10–20	20–30	30–40
Frequency	1	4	3	2

Find the a) Variance b) Skewness c) Kurtosis

***** All The Best *****

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Question 1 [3 × 2 = 6 marks]

1. The probability that a vehicle entering the Luray Caverns has Canadian license plates is 0.12; the probability that it is a camper is 0.28; and the probability that it is a camper with Canadian license plates is 0.09. What is the probability that
 - (a) a camper entering the Luray Caverns has Canadian license plates?
 - (b) a vehicle with Canadian license plates entering the Luray Caverns is a camper?
 - (c) a vehicle entering the Luray Caverns does not have Canadian plates or is not a camper?
2. In a certain region of the country it is known from past experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without cancer as having the disease is 0.06, what is the probability that an adult over 40 years of age is diagnosed as having cancer?

Question 2 [4 × 1 = 4 marks]

Calculate β_1 and β_2 (measures of skewness and kurtosis) for the following frequency distribution and hence comment on the type of the frequency distribution:

x	2	3	4	5	6
f	1	3	7	2	1

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Question 1 [3 × 2 = 6 marks]

- One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?
- Denote by A, B, and C the events that a grand prize is behind doors A, B, and C, respectively. Suppose you randomly picked a door, say A. The game host opened a door, say B, and showed there was no prize behind it. Now the host offers you the option of either staying at the door that you picked (A) or switching to the remaining unopened door (C). Use probability to explain whether you should switch or not.

Question 2 [4 × 1 = 4 marks]

Calculate all the first four moments and skewness, kurtosis for the following frequency distribution and hence comment on the type of the frequency distribution:

Quantity demanded (in units)	60	61	62	63	64	65	66	67	5
Frequency	2	0	15	29	25	12	10	4	3

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Question 1 [3 × 2 = 6 marks]

- The concept of conditional probability has countless uses in both industrial and biomedical applications. Consider an industrial process in the textile industry in which strips of a particular type of cloth are being produced. These strips can be defective in two ways: length and nature of texture. For the case of the latter, the process of identification is very complicated. It is known from historical information on the process that 10% of strips fail the length test, 5% fail the texture test, and only 0.8% fail both tests. If a strip is selected randomly from the process and a quick measurement identifies it as failing the length test, what is the probability that it is texture defective?
- Suppose that the four inspectors at a film factory are supposed to stamp the expiration date on each package of film at the end of the assembly line. John, who stamps 20% of the packages, fails to stamp the expiration date once in every 200 packages; Tom, who stamps 60% of the packages, fails to stamp the expiration date once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp the expiration date once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 200 packages. If a customer complains that her package of film does not show the expiration date, what is the probability that it was inspected by John?

Question 2 [4 × 1 = 4 marks]

Calculate β_1 and β_2 (measures of skewness and kurtosis) for the following survey of 48 engineering companies:

Level of profit (Rs in lakh)	10	11	12	13	14
Number of companies	3	12	18	12	3

***** All The Best *****